

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 12

Aim: Generating correlation matrices using cor() (R).

Code:

```
library(ggplot2)
library(readr)
library(tidyr)
library(dplyr)

house_data <- read_csv("House Price Prediction.csv")
house_subset <- house_data %>%
  select(price, sqft_living, bedrooms, floors, condition) %>%
  head(50) # Using first 50 rows for a clear sample

cor_matrix <- cor(house_subset)

print("Correlation Matrix:")
print(round(cor_matrix, 2))

plot1 <- ggplot(house_subset, aes(x = sqft_living, y = price)) +
  geom_point(color = "darkblue", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  theme_minimal() +
  labs(title = "Scatter Plot: Price vs. Living Space",
       subtitle = "Practical 12: Visualizing Correlation Trend",
       x = "Living Area (sqft)",
       y = "Price ($)")

cor_with_price <- as.data.frame(cor_matrix) %>%
  mutate(Feature = rownames(.)) %>%
  filter(Feature != "price") # Remove the correlation of price with itself (1.0)

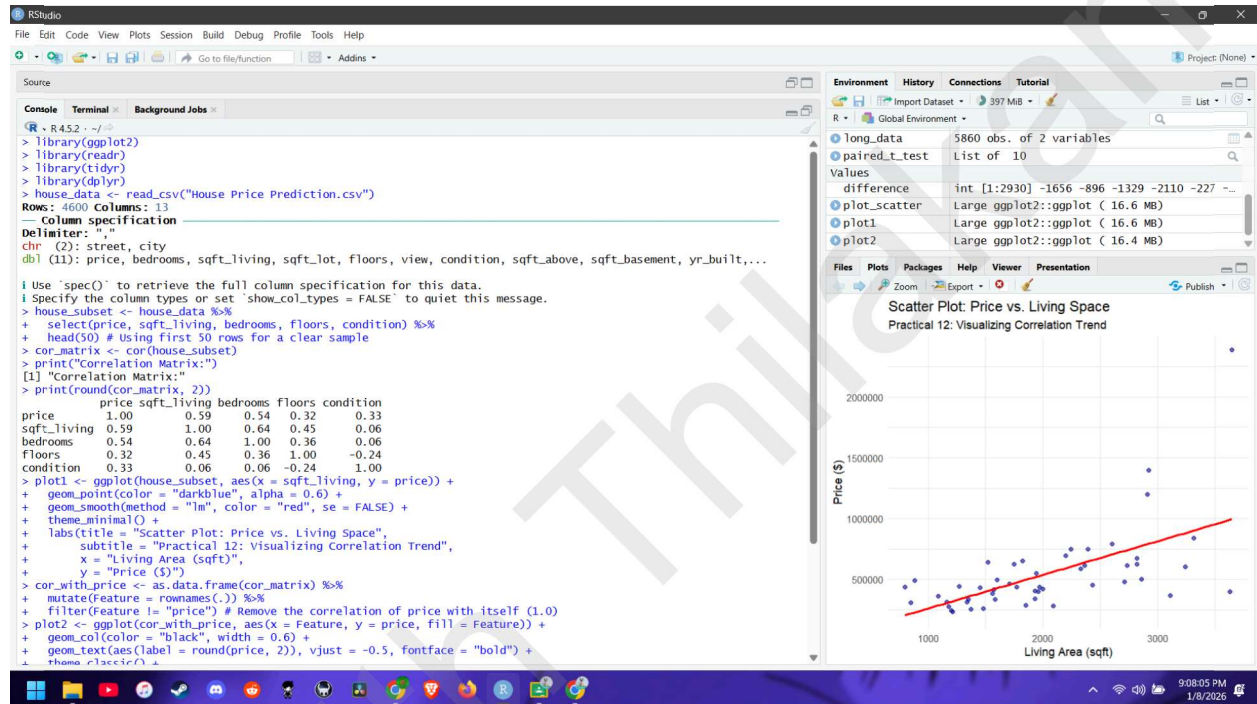
plot2 <- ggplot(cor_with_price, aes(x = Feature, y = price, fill = Feature)) +
  geom_col(color = "black", width = 0.6) +
  geom_text(aes(label = round(price, 2)), vjust = -0.5, fontface = "bold") +
  theme_classic() +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Correlation Strength with House Price",
       subtitle = "Bar representation of correlation coefficients",
```

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```
x = "Feature",  
y = "Correlation Coefficient (r)" +  
theme(legend.position = "none")
```

```
print(plot1)  
print(plot2)
```

Output:



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